NAME OF STUDENT_________________________________SCHOOL_________

MWANZA DISTRICT EXAMINATIONS BOARD

2024 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATIONS

PHYSICS

Subject Number: M164/I

Friday, 22 March **Time Allowed: 2 Hours**

08:00 – 10:00 am

PAPER I

Theory

(100 marks)

Instructions

1. **This paper contains 12 pages. Please check.**
2. Fill in your **Name** at the top of each page.
3. This paper contains **two** sections, **A** and **B**. In **Section A** there are **ten** short answer questions while in **Section B** there are **three** restricted essay questions.
4. Answer all the **thirteen** questions in the spaces provided.
5. Use of electronic calculators is allowed.
6. The maximum number of marks for each question is indicated against each question.
7. In the table provided on this page, **tick** against the number of the question you have answered.

Question Number	**Tick if answered**	**Do not write in these columns**		**Marker's name**
1				
2				
3				
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SECTION A (70 marks)

1a. Mention any **two** factors that affect torque acting on a body

____

____

(2 marks)

b. What type of waves are electromagnetic waves?

____

(1 mark)

c. Give any two applications of infrared waves.

____

____

(2 marks)

d. A convex lens forms an image whose magnification is $-\mathbf{2}$. Describe the nature of this image.

____

____

(2 marks)

2a. Why is the focal length in a camera fixed?

____

(1 mark)

b. Give the **two** conditions for the occurrence of total destructive interference in transverse waves.

____

____

(2 marks)

c. State how the amplitude of an incident sound wave front is determined.

____

(1 mark)

d. **Figure 1** shows two identical bottles containing water as shown.

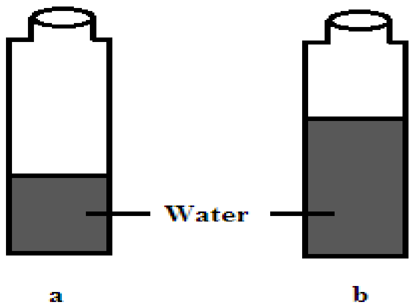

Figure 1

(i) In which bottle will the sound have a greatest pitch?

____

(1 mark)

(ii) Explain the answer to 2d (i) above

____

____

(2 marks)

3a. Name the type of thermometer that is used as a flame sensor in safety devises.

____

(1 mark)

b. **$4m^3$** of a fixed mass of argon gas initially at **770mmHg** is compressed to half its original volume at constant temperature. Calculate the final pressure of the gas.

(4 marks)

c. Differentiate absolute error from relative error

____

____

(1 mark)

4a. The half-life of a certain radioisotope is 28 days.

(i) What does this mean?

____

(1 mark)

(ii) What will happen to the value of this half-life if the radioisotope of this type was allowed to decay in an environment of a higher temperature?

____

(1 mark)

(iii) Give your answer to 4(ii) above.

____

____

(1 mark)

b. **Figure 2** shows radiation passing through an electric field between two plates **R** and **S**.

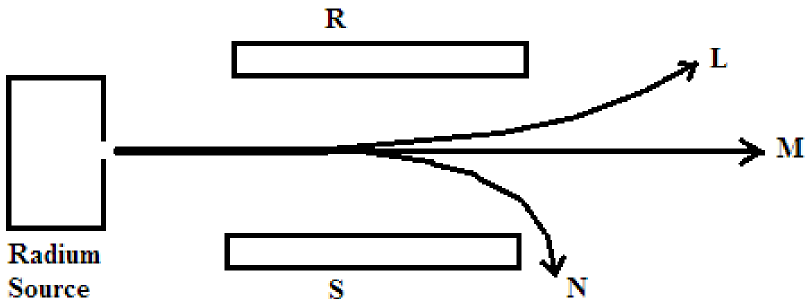

Figure 2

(i) Identify the charge in plate S in the **Figure 2**.

____

(1 mark)

(ii) Explain the answer to 4b (i) above.

____

____

(2 marks)

c. State how controlled chain reaction can be achieved in a nuclear reactor?

____

(1 mark)

5a. **Figure 3** shows a circuit diagram with the three light bulbs as shown.

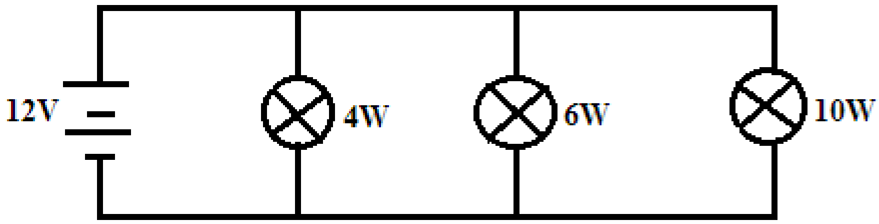

Figure 3

(i) What does the rating of **4W** mean?

____

(1 mark)

(ii) What would be the effect on the power dissipated by the **10W** bulb if the **6W** bulb burns out?

____

(1 mark)

(iii) Give two reasons for the answer to 5a(ii) above.

____

____

(2 marks)

b. Mention the two semiconductor electronic components that store energy.

____

____

(2 marks)

c. State the functional difference between a light emitting diode and a photovoltaic cell.

____

____

(1 mark)

6.a Name the law that is used to determine the direction of induced emf when a bar magnet is moved into a fixed conductor.

____

(1 mark)

b. State the advantage of using a zener diode as a safety devise over using a fuse in an electric circuit.

____

(1 mark)

c. Construct a truth table for the operation of a NOR logic gate.

(4 marks)

d. How does the magnitude of coefficient of static friction compare with that of a dynamic friction?

____

(1 mark)

7a. How much effort is required to lift a 200N load using a pulley system with four pulleys?

(2 marks)

b. Describe how clothes dry up in an electric drier.

____

____

____

(3 marks)

c. **Figure 4** shows two bodies, A and B revolving around a fixed force, Q, along their own orbits.

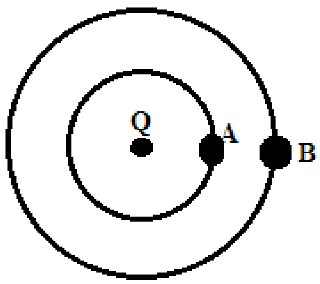

Figure 4

(i) How does the tangential velocity of body **A** compare with that of body **B** if they both take same time to complete a revolution?

____

(1 marks)

(ii) Give a reason for the answer to 6c (i) above.

____

(1 mark)

8a. Calculate mass, in Kg, that can produce an extension of 5cm when two identical springs of spring constant 50N/cm each are arranged in parallel.

(3 marks)

b. Give two differences between elastic collision and inelastic collision

____

____

(2 marks)

c. Why is a body moving in a uniform circular motion said to accelerate despite moving at a constant speed?

____

(1 mark)

9a. Describe how a crumple zone of a car provides safety to passengers in a car.

____

____

____

(2 marks)

b. **Figure 5** shows a 10 kg block lying on a flat, rough horizontal surface of a table. The block is connected to a 2 kg block hanging over the side of the table. The string runs over a light frictionless pulley. The blocks are stationary at the point when they are just about to move.

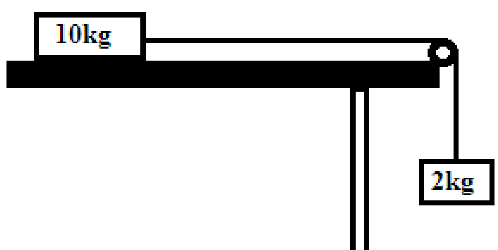

Figure 5

(i) Calculate coefficient of static friction force for the 10kg block.

(3 marks)

(ii). What would happen to the value obtained in 9b (i)above if the mass of the 10kg block was increased?

____

(1 mark)

(iii). Give a reason for the answer in 9b(ii) above.

____

(1 mark)

10a. Construct a fixed resistor using resistor color codes with the value of 25 k$\Omega \pm 10\%$

(3 mark)

b. State the energy-work theorem

____

____

(1 mark)

c. Give two factors that affect power loss in cables during transmission of electrical power.

____

____

(**2 marks**)

d. State the meaning of phase of a wave

____

(1 mark)

SECTION B (30 marks)

11a. Describe how a U – tube manometer works to measure gas pressure

(5 marks)

b. Describe how fish survive in a freezing water body in cold regions.

(5 marks)

12a. Briefly describe how p-type semiconductors are produced.

(5 marks)

12b. With the aid of a well labeled diagram, describe the nature of reflected waves when straight wave fronts are allowed to strike a convex reflecting surface.

(5 marks)

13. With the aid of a well labeled graph, describe the motion of a body as it falls through a liquid.

(10 marks)

END OF QUESTION PAPER

NB: This paper contains 12 printed pages.